E-Waste & Environmental Management

Activity Worksheet

The Hidden Carbon Footprint of Our Gadgets

Name: Swapnil Bhabal	**Roll No.:** 24101B0065
Division: B	**Date:** 5/8/2026

Nice. 9 → 10. Nice.

Part A: Carbon Footprint Calculation

Use the carbon footprint calculator provided by your faculty.

Parameter	Your Response
Annual Carbon Footprint	11 tCO_2e / year
National Average	7 tCO_2e / year
Global Average	19 tCO_2e / year
Number of Earths Required (if provided)	0.58

Please fix.

Which category contributed the most?

☐ Transportation ☐ Home Energy ☐ Food ☑ Shopping & Consumption

☐ Electronics / Digital Lifestyle ☐ Other: _____

Which result surprised you the most?

I was ~~super~~ suprised that my footprint (11 tons) is higher than the Indian national average (7 tons), even though it is still much lower than the global average (19 tons)

Part B: My Electronic Lifestyle

1. How many smartphones have you owned in the last 10 years?

☑ 1 ☐ 2–3 ☐ 4–5 ☐ More than 5

2. How often do you replace your smartphone?

☐ Every year ☐ Every 2 years ☐ Every 3–4 years ☑ More than 5 years

3. What do you usually do with your old phone?

☑ Keep ☐ Sell ☐ Exchange ☐ Donate

☐ Recycle ☐ Throw away

4. Have you ever repaired an electronic device instead of replacing it?

☑ Yes ☐ No

If yes: Fixed a cracked phone screen

5. Have you ever recycled e-waste through an authorized recycler?

☐ Yes ☑ No

Item: _____

Unused Electronic Items at Home	Number
Mobile phones	3
Chargers	2
Earphones	1
Batteries	0
Cables	3
Other	1

Part C: Think Like an Environmental Engineer

Habit	Higher Carbon Footprint	More E-Waste
Buying a new phone every 2 years	☑	☑
Repairing a laptop	☐	☐
Keeping old phones unused in drawers	☐	☑
Recycling electronic devices	☐	☐
Using devices for 5–6 years	☐	☐

Which stage of an electronic product's life cycle contributes the MOST carbon emissions?

☐ Raw material extraction ☑ Manufacturing ☐ Transportation

☐ Usage ☐ Disposal

Why is extending the life of an electronic device environmentally beneficial?

Extending the life of an electronic device reduces the need to manufacture new products, conserving raw materials and energy, lowering carbon emissions and reducing electronic waste (e-waste)

Part D: Reflection

Three habits that increase my personal carbon footprint:

1] Leaving electrical appliances, laptops, lights and chargers plugged in or running when not actively in use
2] Purchasing unnecessary products and accessories
3] Using electronic devices for long hours everyday

Three habits that increase e-waste generation:

1] Storing old, non-working cables, earphones and chargers in household drawers instead of properly recycling them
2] Disposing of small consumer electronics and dead lithium batteries into the regular household garbage bin
3] Buying low-quality electronic accessories that fail quickly

Action to Reduce Carbon Emissions & E-waste	Expected Impact
Use my smartphone for 5-6 years before replacing it	Reduces carbon emissions and e-waste
Repair devices instead of buying new ones	Extends device life and save resources
Recycle or donate old electronic devices	Prevents e-waste and recovers valuable materials
Use energy efficient devices and switch them off when not in use	Reduces electricity consumption, lowers carbon emissions.

Discussion Questions (Optional)

1. Which stage of an electronic product's life cycle has the highest carbon footprint and why?

The manufacturing stage has the highest carbon footprint because it requires mining raw materials, producing components and assembling the device which consume large amounts of energy & generated high carbon emissions

2. How does extending the lifespan of smartphones reduce both carbon emissions and e-waste?

Using smartphones for a longer time reduces the need to manufacture new phones, which lowers carbon emissions. It also delays disposal, reducing the amount of e-waste generated.

3. Why is recycling alone not enough for sustainable e-waste management?

Recycling is important but it cannot recover all materials and still requires energy. Reducing, reusing, repairing and refurbishing electronic devices are more effective ways to minimize environmental impact

4. Explain how the 5Rs contribute to lowering the environmental impact of electronic products.

The 5Rs - Refuse, Reduce, Reuse, Repair and Recycle help lower environmental impact by reducing unnecessary purchases, extending the life of devices, conserving natural resources, decreasing carbon emissions and minimizing e-waste.

5. As an engineering student, what innovative solution would you propose to reduce the carbon footprint associated with electronic devices?

I would develop a smart e-waste management app that encourages people to repair, reuse and recycle electronic devices. It would locate nearby e-wast collection centers and reward user for responsible disposal, helping reduce both carbon emission and e-waste

Ms. Nilima Main
Assistant Professor, Department of First Year Engineering
Vidyalankar Institute of Technology